install Cyclictest:

```
sudo apt-get install rt-tests
```

Once installed, run Cyclictest with distinct options. The basic syntax is as follows:

```
cyclictest <options>
```

Here are some common options used with Cyclictest:

-t: Specifies the number of threads to use (default is 1)

-n: Specifies the number of iterations (default is 10000)

-i: Specifies the interval (in microseconds) between iterations (default is 1000µs)

-l: Specifies the maximum thread lock latency (in microseconds) before considering a result as an outlier (default is 10000µs)

-m: Specifies the CPU mask, which restricts the test to specific CPUs (default is all CPUs)

-p: Prints the raw latency results

-H: Prints the latency histogram

-S: Prints the latency statistics (mean, min, max, etc)

-q: Quiet mode; suppresses the summary and histogram outputs.

Here is a Cyclictest sample command:

```
cyclictest -p 90 -t80 -m -n 10000 -l
100000 -priospread
```

This command uses the values given below with options.

- The option *-p* is used for thread priority level and *-p* as 90.
- The option *–t* is used for number of threads and *-t* as 80.
- The option *-m* is used to enable *lockall*, which locks all memory used by the 'cyclictest' process into RAM.
- Number of iterations or number of test cycles *-n* is 10000.
- The option *-l* sets the length of each test cycle in microseconds and *-l* as 100000.
- The option *-priospread* is used to enable 'priority spreading'. Priority spreading is a feature in the Linux kernel that helps mitigate priority inversion issues.

Kernel configurations

The kernel configurations given below enable the PREEMPT_RT patch. The latencies are calculated with all the kernel configurations as well as with the enabled PREEMPT_RT patch. The testing is done with 'cyclictest' Linux command-line utility with some required options. Hackbench load is also given with the Cyclictest command.

The kernel config options (including PREEMPT_RT Patch) are:

- `Kernel version: 6.1.35`
- `Hardware: Raspberry Pi 4 model B (64-bit)`

Kernel configuration options for the server should be 'No Forced Preemption' as shown in Figure 2.

These changes are reflected in the *.config* file as shown below:

```
CONFIG_PREEMPT_NONE_BUILD=y
CONFIG_PREEMPT_NONE=y
CONFIG_PREEMPT_VOLUNTARY is not set.
CONFIG_PREEMPT is not set.
CONFIG_PREEMPT_DYNAMIC is not set.
CONFIG_SCHED_CORE is not set.
```

Kernel configuration options for the desktop should be 'Voluntary Kernel

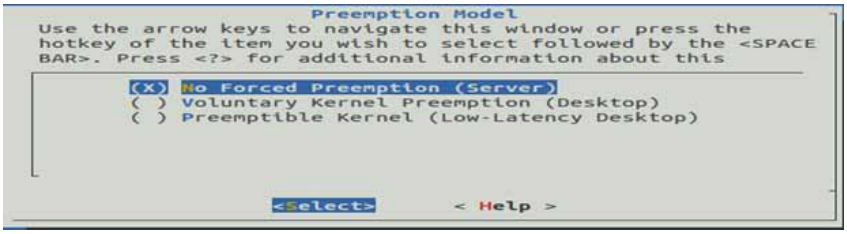

Figure 2: Kernel config options for server – 'No Forced Preemption'

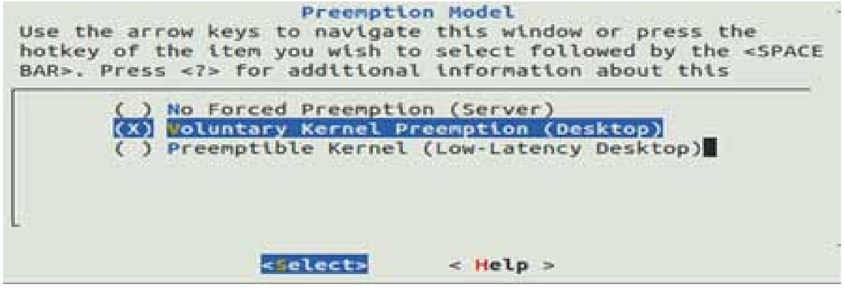

Figure 3: Kernel config options for the desktop – 'Voluntary Kernel Preemption'

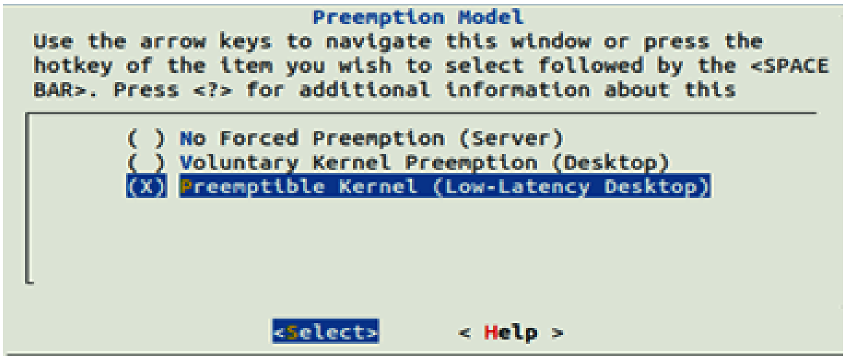

Figure 4: Kernel config options – 'Preemptible Kernel'